

Human-AI Collaboration in Scientific Writing and Research Workflows: A Comprehensive Survey

Abstract

The integration of Generative Artificial Intelligence (GenAI) into scientific research has precipitated a paradigm shift from static tool usage to dynamic, agentic collaboration. This survey synthesizes findings from 300 recent academic studies to map the evolving landscape of Human-AI collaboration in scientific writing and research workflows. We categorize the field into distinct methodological families, ranging from assistive co-writing interfaces and retrieval-augmented literature synthesis to fully autonomous “AI Scientist” agents capable of end-to-end hypothesis generation, experimentation, and manuscript drafting. Our analysis reveals a critical tension: while AI significantly accelerates ideation, coding, and drafting phases—particularly benefiting non-native English speakers and smaller research teams—it introduces systemic risks regarding hallucination, verification bottlenecks, and the erosion of authorial agency. We examine the technical architectures enabling these systems, including multi-agent orchestration, model context protocols, and evidence-gated control planes. Furthermore, we critically evaluate the socio-technical implications, including the “denominator gaming” of peer review, the homogenization of scientific discourse, and the urgent need for new infrastructure to support process-oriented provenance. This survey concludes that the future of scientific discovery lies not in full automation, but in “mixed-initiative” ecosystems where humans transition from executors to curators and strategic directors of AI-driven research loops.

1. Introduction

The scientific method, historically a human-centric endeavor involving hypothesis formulation, experimental validation, and peer dissemination, is undergoing a fundamental transformation driven by Large Language Models (LLMs) and agentic AI systems. The emergence of generative AI has moved beyond simple text completion to encompass complex research workflows, including literature review automation, code generation, experimental design, and even autonomous peer review [1], [2], [3]. This evolution marks a transition from AI as a passive tool to AI as an active collaborator, fundamentally altering the cognitive load, temporal dynamics, and structural organization of scientific teams [4], [5].

Recent bibliometric analyses indicate an exponential surge in AI-assisted publications, with GenAI adoption correlating with increased productivity, particularly among non-native English speakers and early-career researchers [6], [7], [8]. Studies show that AI-assisted teams are becoming smaller and younger without compromising scientific impact, suggesting a democratization of research capabilities [4]. However, this acceleration brings profound challenges. The “verification burden” has shifted from data collection to the auditing of AI-generated content, creating a potential bottleneck where writing speed outpaces the capacity for rigorous validation [9], [10]. Furthermore, the stochastic nature of LLMs introduces risks of hallucination, citation fabrication, and the propagation of “tortured phrases” that threaten the integrity of the scientific record [11], [12], [13].

This survey provides a comprehensive synthesis of the current state of Human-AI collaboration in scientific writing and research. We organize the field into a coherent taxonomy based on the level of autonomy and the specific stage of the research lifecycle addressed. We explore methodological advancements in tool-augmented reasoning, multi-agent systems, and retrieval-augmented generation (RAG) that underpin modern research assistants [14], [15], [16]. We further analyze the

socio-technical dimensions, including the psychological impact on authorship, the ethical implications of undisclosed AI use, and the emerging policy frameworks designed to govern this new era of “vibe researching” and autonomous discovery [17], [18], [19]. By synthesizing evidence from 300 distinct studies, we aim to provide a rigorous foundation for understanding the opportunities and perils of AI-integrated science.

2. Method

The methodological landscape of Human-AI collaboration in science is diverse, spanning from lightweight writing assistants to complex, closed-loop autonomous agents. We categorize these approaches into several interconnected families: collaborative writing interfaces, automated literature synthesis, agentic research workflows, autonomous scientific discovery, and infrastructure for reproducibility and governance.

2.1. Collaborative Writing Interfaces and Cognitive Scaffolding

The most immediate impact of GenAI in science is observed in the writing process, where systems are designed to augment human cognition rather than replace it. These systems vary in their level of scaffolding, from sentence-level completion to hierarchical document planning.

2.1.1. Scaffolding Levels and User Agency Research indicates that the granularity of AI assistance significantly impacts writing quality and user ownership. High-scaffolding systems, which provide paragraph-level suggestions, have been shown to boost writing quality and productivity, particularly for non-regular writers, whereas low-scaffolding sentence completions can sometimes decrease quality or lead to “mindless echoing” [20], [21]. The concept of “reactive writing” suggests that inline suggestions interrupt internal ideation, forcing a shift from generation to evaluation, which can alter the writer’s original argumentative stance [22]. To mitigate the loss of psychological ownership, recent interfaces employ friction-based nudges, such as time delays inversely proportional to prompt length, encouraging deeper cognitive investment [23]. Furthermore, style-personalized suggestions that condition outputs on the user’s historical writing voice have been found to partially restore feelings of ownership compared to generic suggestions [24].

Advanced interfaces are moving beyond linear text editing to support non-linear cognitive processes. Systems like **TreeWriter** represent documents as hierarchical trees, separating idea structure from final text to reduce cognitive load and enable progressive disclosure [25]. Similarly, **VISAR** utilizes visual programming to synchronize text blocks with argumentative graphs, allowing writers to maintain global logical coherence while refining local content [26]. The **CoAuthor** dataset and related studies highlight that while LLMs generate grammatically superior text, they often lack vocabulary diversity, necessitating human intervention to maintain creative variation [27], [28].

2.1.2. Domain-Specific Writing Assistance In specialized domains, generic LLMs often fail to meet rigorous standards. **Bibby AI**, a native LaTeX editor, integrates AI directly into the document’s Abstract Syntax Tree (AST), enabling real-time error detection and fix generation that outperforms external extensions [29]. For regulatory writing, such as Investigational New Drug (IND) applications, systems like **AutoIND** utilize domain-specific templates to reduce drafting time by over 96%, though they require aggressive human editing to correct verbosity and omissions [30]. In the realm of figure captions, tools like **SciCarpenter** provide contextual checklists and ratings alongside generated text, reducing cognitive load more effectively than raw generation [31].

Studies on astrophysics workflows reveal that while AI assists well with creative and extractive tasks, it can be catastrophic for derivation-heavy reasoning if not carefully constrained [32].

2.1.3. Co-Writing Dynamics and Feedback Loops The interaction dynamics between humans and AI are critical. **CounterQuill** demonstrates that structured educational workflows, which decompose complex communication tasks into learning, brainstorming, and co-writing phases, enhance user confidence and skill acquisition more effectively than direct generation [33]. In educational settings, theory-driven Learning Analytics Dashboards (LADs) grounded in Self-Regulated Learning (SRL) frameworks have been shown to improve writing knowledge gains by transforming abstract rules into actionable feedback [34]. However, the “dearth of the author” framework warns that delegating creative decisions to AI can lead to stylistic homogenization and reduced authorial intent, advocating for tools that preserve the ratio of expressive intent to output length [35].

2.2. Automated Literature Synthesis and Knowledge Discovery

Literature review and synthesis represent a major bottleneck in research, prompting the development of sophisticated AI systems capable of retrieving, clustering, and summarizing vast corpora.

2.2.1. Retrieval-Augmented Generation (RAG) and Survey Generation Automated survey generation has evolved from naive RAG pipelines to complex, multi-stage workflows. **AutoSurvey** and **AutoSurvey2** employ hierarchical parallel drafting and graph-based state machines to ensure structural coherence and citation fidelity, outperforming baselines in coverage and relevance [36], [37]. **SurveyGen-I** introduces dynamic planning mechanisms (PlanEvo) that refine outlines based on accumulated writing memory, mitigating structural fragmentation [16]. More advanced systems like **ARISE** utilize rubric-guided iterative refinement with specialized agents for citation preparation and knowledge base construction, achieving high scores in expert evaluations [38]. **MVSS** adopts a structure-first paradigm, jointly optimizing hierarchical knowledge trees and comparison tables to superior conceptual alignment [39].

To address the issue of citation hallucination, systems like **LiRA** decompose review generation into specialized agents (Outline Drafter, Subsection Writer, Editor) with explicit citation grounding via full-article retrieval [40]. **TOPICAL** employs centroid-first sampling strategies to ensure diverse context selection within token limits, while mandating inline PMID citations for verifiability [41]. **LitLLM** integrates external academic search APIs with LLM-based re-ranking to ground generation in factual content, reducing hallucination rates [42].

2.2.2. Interactive Synthesis and Sensemaking Moving beyond static generation, interactive systems facilitate human-AI sensemaking. **Synergi** constructs local citation graphs and uses Loopy Belief Propagation to rank candidate papers, creating boundary objects that bridge manual discovery and automated synthesis [43]. **AwesomeLit** visualizes agent reasoning as a controllable node-link flow, allowing users to intervene at specific nodes (Search, Review, Synthesis) to correct errors without restarting sessions [44]. **DimInd** employs a sequential transformation pipeline, progressively disclosing structured representations from tables to taxonomies to narratives, which serves as an external cognitive scaffold to reduce navigation costs [45]. **PaperBridge** utilizes bi-directional analysis engines to scaffold research narratives, combining framework-guided top-down suggestions with user-driven bottom-up refinement [46].

2.2.3. Data Extraction and Table Generation Extracting structured data from unstructured literature is another critical application. **ARXIVDIGESTABLES** decomposes table generation into schema induction and value extraction phases, leveraging rich contextual cues like captions to improve reconstruction recall [47]. **SciDaSynth** integrates RAG with interactive visual summarization to resolve data inconsistencies across documents, enabling efficient semantic grouping of extracted values [48]. **DeepShovel** implements a modular human-AI collaboration framework for geoscience literature, separating extraction tasks (meta, text, table, map) and allowing users to verify AI proposals iteratively [49].

2.3. Agentic Research Workflows and Tool Use

The integration of AI agents capable of planning, tool use, and iterative refinement marks a significant advancement over static text generation. These systems orchestrate complex workflows involving code execution, data analysis, and external API calls.

2.3.1. Multi-Agent Orchestration and Planning Multi-agent systems (MAS) are increasingly used to simulate research teams. **AISSISTANT** employs a dual-workflow architecture with specialized agents for research and paper writing, utilizing a formal asset propagation mechanism to enable selective human intervention [14]. **PaperOrchestra** executes a five-stage pipeline with parallel agents for outlining, plotting, and literature review, using hybrid discovery to prevent hallucination [50]. **Story2Proposal** coordinates architect, writer, refiner, and renderer agents via a persistent shared visual contract to maintain global coherence [51]. **Cocoa** embeds AI agents into document editors via interactive plans, enabling fluid transitions between planning and execution phases [52].

Advanced orchestration frameworks like **ResearchLoop** define persistent state objects and evidence-gated claim admission to prevent unsupported assertions, externalizing claim ledgers to versioned repositories [53]. **Human Tool** reframes humans as episodic callable resources within AI-led workflows, shifting orchestration responsibility to the AI to reduce cognitive load [54]. **VAILabs** employs a modular architecture with XML-defined workflows, supporting conditional loops and human checkpoints for cross-domain reuse [55].

2.3.2. Tool Learning and Function Calling Effective agent behavior relies on robust tool use. **Paper2Agent** transforms static research papers into interactive Model Context Protocol (MCP) servers, encapsulating methods as executable tools that LLM agents can invoke reliably [56]. **TIB AIssistant** integrates curated prompt libraries with external scholarly tools via MCP, lowering barriers to adoption while maintaining researcher agency [57], [58]. **Agent Laboratory** utilizes an **mle-solver** module for iterative ML code generation and a **paper-solver** for automated peer review, demonstrating that human involvement in “Co-pilot mode” significantly improves quality metrics [59].

In mathematics and coding, **The Agentic Researcher** wraps general-purpose CLI agents in sandboxed containers governed by “Ten Commandments” of scientific rigor, enforcing single-variable experimentation and tiered evaluation [60]. **Ten Simple Rules for AI-Assisted Coding** emphasizes the separation of problem framing from implementation syntax, advocating for test-driven development to prevent “paper tests” [61]. **Re-opening open-source science** demonstrates that segregating high-reasoning planning tasks from fast coding tasks under human architectural guidance enables robust modification of large legacy codebases [62].

2.3.3. Iterative Refinement and Self-Healing Autonomous agents often struggle with long-horizon tasks due to error accumulation. **AutoResearchClaw** implements self-healing executors utilizing Pivot/Refine loops to recover from code failures, transforming execution errors into diagnostic information [63]. **Omakase** monitors user project documents to generate timely, actionable suggestions, discarding inactive threads to prevent stale recommendations [64]. **PromptPilot** implements an interactive feedback loop that identifies error domains and generates goal-oriented refinement suggestions, bridging the gap between static guidelines and dynamic task performance [65].

2.4. Autonomous Scientific Discovery (“AI Scientists”)

The frontier of this field involves fully autonomous systems capable of executing the entire research lifecycle, from ideation to publication. These “AI Scientists” represent a shift from assistance to autonomy.

2.4.1. End-to-End Research Loops Systems like **The AI Scientist** execute a three-phase loop: idea generation via LLM mutation, experiment iteration using coding agents, and paper write-up, achieving balanced accuracy comparable to human baselines in specific domains [66]. **Jr. AI Scientist** mimics novice researcher workflows by analyzing baseline papers, generating improvement hypotheses, and iterating experiments until performance gains are achieved, though it tends to produce incremental rather than novel contributions [67]. **AI-Researcher** orchestrates a three-stage pipeline for literature review, idea generation, and implementation, leveraging bidirectional mapping between mathematical formulations and code [68].

More complex systems like **OmniScientist** integrate dynamic knowledge graphs and network-augmented BFS for literature review, enabling agents to transition from tools to collaborators by encoding human research infrastructure [69]. **Virtuous Machines** coordinates hypothesis generation, experimental design, and real-world implementation via online participant platforms, completing end-to-end studies in hours rather than months [70]. **PARNES** implements a declarative DAG kernel with a four-field agent contract, allowing dynamic, discipline-specific loops and cross-run knowledge accumulation [71].

2.4.2. Hypothesis Generation and Novelty Assessment Generating novel and feasible hypotheses remains a core challenge. **HARPA** employs a divergence-convergence generator with a reward-based scorer, using Socratic questioning to refine hypothesis specificity against literature snippets [72]. **Scideator** extracts purpose, mechanism, and evaluation facets from papers to recombine them into novel research ideas, utilizing a two-stage re-ranking system for novelty assessment [73]. **HybridQuestion** integrates AI-accelerated information gathering with cross-model ensemble voting to identify high-impact research questions, showing high alignment with humans on retrospective breakthroughs [74]. **IRIS** adapts Monte Carlo Tree Search (MCTS) for subjective scientific ideation, using LLM-based proxy judges to navigate the idea space systematically [75].

However, evaluations reveal significant limitations. **EXP-Bench** shows that while agents achieve partial success in design or implementation individually, they collapse under conjunctive requirements, with top agents scoring only 1.4% on strict correctness metrics [76]. **AI Scientists Fail Without Strong Implementation Capability** highlights that LLMs exhibit severe performance degradation in complex engineering tasks compared to simple generation, with result match accuracy as low as 0.7% on rigorous benchmarks [77].

2.4.3. Automated Experimentation and Robotics Physical automation extends AI discovery into the lab. **Airalogy** couples executable protocols with structured data records, utilizing LLM-agents as universal adapters to bridge domain expertise and computational standardization [78]. **Scaling Laws in Scientific Discovery** integrates agentic AI with embodied robotics, using OS agents to navigate digital environments and general-purpose robots for physical manipulation [79]. **Accelerating Discovery in Natural Science Laboratories** proposes a symbiotic human-machine feedback loop where machines handle robustness constraints while humans provide expert interpretation [80]. Despite progress, full autonomy remains impractical due to hardware engineering costs and the fragility of modular workflows [81], [80].

2.5. Evaluation, Reproducibility, and Governance

As AI-generated content proliferates, new methods for evaluation, reproducibility, and governance are essential to maintain scientific integrity.

2.5.1. Automated Peer Review and Quality Assessment AI is increasingly used to augment or automate peer review. **ReviewRobot** constructs knowledge graphs to compare target papers against background literature, achieving recommendation accuracy close to human inter-annotator agreement [82]. **GUIDE** utilizes modular section-wise summarization and rubric-guided prompts to evaluate novelty and significance, outperforming large general-purpose models in acceptance rate prediction [83]. **Agents4Science** implemented a dual-track review system where LLMs scored submissions, revealing that higher-quality accepted papers correlated with increased human involvement in early stages [84]. However, concerns persist regarding “denominator gaming,” where automated agents inflate submission volumes to exploit fixed acceptance rates, potentially lowering quality thresholds [85].

2.5.2. Reproducibility and Artifact Generation Ensuring reproducibility is critical. **AI Copilots for Reproducibility** generate structured Jupyter Notebooks with placeholders for missing logic, reducing reproduction time by 97% [86]. **Scaling Reproducibility** decouples scientific reasoning from computational execution via three-layer agentic orchestration, enabling reliable scaling of reproducibility checks [87]. **The Last Human-Written Paper** proposes Agent-Native Research Artifacts (ARA) that replace narrative papers with structured file-system ontologies, preserving branching research trajectories and dead-end nodes to reduce redundant exploration [88]. **ClawXiv** implements a signed archival workflow, decoupling mutable project directories from immutable signed bundles to ensure reproducibility [89].

2.5.3. Detection, Disclosure, and Ethics Detecting AI use and ensuring transparency are active areas of research. While detectors like **Originality.AI** claim high accuracy, their performance degrades in fields with high symbolic content, and they often yield false positives on translated texts [90], [13]. Consequently, policy frameworks emphasize disclosure over detection. **PaperCard** and the **Artificial Intelligence Disclosure (AID)** framework propose structured metadata and granular taxonomies to map AI roles from conceptualization to administration [91], [92], [19]. **PAIRED** shifts documentation from research products to decision points, capturing epistemically distinct collaboration modes [93]. Despite these efforts, studies show that journal policies have failed to curb the surge in AI-assisted writing, and reviewers often cannot reliably distinguish AI-generated abstracts from human-written ones [94], [95].

3. Challenges and Outlook

Despite rapid advancements, the integration of AI into scientific workflows faces significant technical, ethical, and systemic challenges.

3.1. The Verification Bottleneck and Hallucination

A primary concern is the asymmetry between the speed of AI-generated content and the capacity for human verification. Dynamical models suggest that as writing speedup exceeds review speedup, a “verification debt” accumulates, potentially leading to a degradation of scientific knowledge [9]. Hallucinations remain pervasive, particularly in citations and experimental details, with some systems fabricating non-existent references with high confidence [11], [96], [12]. While retrieval-augmented generation mitigates this, it does not eliminate the risk of “context-induced bias” or the fabrication of datasets in autonomous agents [97], [98]. The “implementation gap” further exacerbates this, where agents can propose plausible ideas but fail to execute the complex code required to validate them [77], [76].

3.2. Authorship, Agency, and Skill Erosion

The shift from “researcher-as-creator” to “researcher-as-curator” raises profound questions about authorship and skill retention. Studies indicate that high levels of AI assistance can reduce psychological ownership and lead to the “dearth of the author,” where creative decision-making is disproportionately low [35], [99]. There is a risk of “skill erosion,” where over-reliance on AI for coding and reasoning leads to the atrophy of first-principles thinking essential for validating novel results [18], [100]. Furthermore, the “reactive writing” phenomenon suggests that constant AI suggestions may alter the cognitive process of ideation, potentially narrowing the diversity of scientific thought [22], [101].

3.3. Equity, Bias, and Homogenization

While AI acts as a linguistic equalizer for non-native English speakers, it also risks homogenizing scientific discourse. Large-scale analyses reveal that LLMs induce consistent semantic shifts toward a shared region in embedding space, reducing lexical diversity and potentially creating a “monoculture” of scientific writing [101], [8]. Additionally, context-induced bias can trigger stereotype-aligned responses based on user demographics, and the reliance on open-access literature excludes negative results and paywalled data, skewing the knowledge base [97], [102]. The “Matthew Effect” may be exacerbated if AI tools prioritize highly cited but potentially outdated sources, reinforcing existing biases in the literature [103].

3.4. Infrastructure and Policy Gaps

Current academic infrastructure is ill-equipped for the AI era. Traditional publication formats (PDFs) compress argument graphs into linear narratives, making verification difficult [104]. There is a lack of standardized benchmarks for evaluating autonomous agents, with most systems tested only in narrow domains like machine learning [15], [105]. Policy frameworks are heterogeneous and often ineffective, with many journals lacking clear guidelines or enforcement mechanisms [94], [106]. The rise of “agentic denominator gaming” threatens the integrity of peer review, necessitating new governance models that go beyond simple disclosure [85]. Future research must focus on developing “inspectable AI” frameworks that treat AI interactions as research objects with full provenance

tracking [107], and on creating “living reviews” that can be dynamically updated by AI agents [108].

4. Conclusion

The integration of Artificial Intelligence into scientific writing and research workflows represents a transformative epoch in the history of science. As evidenced by the 300 studies surveyed, AI has evolved from a passive writing aid to an active collaborator capable of orchestrating complex research loops, generating hypotheses, and even conducting autonomous experiments. Methodological innovations in multi-agent systems, retrieval-augmented generation, and interactive interfaces have significantly lowered barriers to entry, accelerated discovery cycles, and enhanced the productivity of diverse research teams.

However, this transformation is not without peril. The field stands at a critical juncture where the efficiency gains of AI must be balanced against the imperatives of rigor, reproducibility, and human agency. The risks of hallucination, verification bottlenecks, and the potential erosion of deep scientific skills demand a rethinking of our research infrastructure and ethical frameworks. The future of scientific discovery will likely not be fully autonomous but rather a “mixed-initiative” ecosystem. In this vision, humans assume the role of strategic directors and curators, leveraging AI to handle routine execution and data synthesis while retaining oversight over hypothesis validation, ethical judgment, and creative direction. Realizing this potential requires concerted efforts to develop robust evaluation benchmarks, transparent provenance systems, and adaptive policies that foster responsible innovation while safeguarding the integrity of the scientific record.

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